

Stabilizing Causal Estimates: Tackling Random Seed Dependence in Machine Learning

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¹Footnotes are for asides or references, but fair game for questions

Steps for Causal Inference

Identification

- Translate parameter into observed data
- Machine learning unlikely to be helpful here

Estimation

- Use observed data expression to estimate the parameter
- Machine learning may be useful²

²Machine learning is not magic. Even data-adaptive algorithms can need variable pre-processing or transformations for validity. See Naimi, Mishler, & Kennedy *American Journal of Epidemiology* 2023

Notation

Y^a : potential outcome under action a

$Z = (W, A, Y)$

Y : outcome of interest

A : action of interest

W : covariate vector

- $\mathcal{W}$ denotes the support

Parameter: average causal effect

$$\psi = E[Y^1 - Y^0]$$

Express parameter of interest as a function of the observed data

Given causal consistency, exchangeability by W with positivity

$$\psi = E [E(Y \mid A = 1, W)] - E [E(Y \mid A = 0, W)]$$

Estimation³

AIPW estimator

$$\hat{E}[Y^a] = \frac{1}{n} \sum_{i=1}^n \left\{ \mu(a, W_i) + \frac{I(A_i = a) [Y_i - \mu(a, W_i)]}{\pi(W_i)} \right\}$$

where

$$\pi(W) = \Pr(A = a \mid W)$$

$$\mu(A, W) = E(Y \mid A, W)$$

as π and μ are unknown in observational research, need to be estimated instead

³Not all identified parameters are estimable, see Maclaren & Nicholson 2021 or Aronow et al. 2025 *Observational Studies*

Dimension of $\mathcal{W}$

Let p denote the size of $\mathcal{W}$

When $n > p$ can fit a saturated model for π or μ .⁴

- No modeling constraints imposed

Often that $n \ll p$ in practice

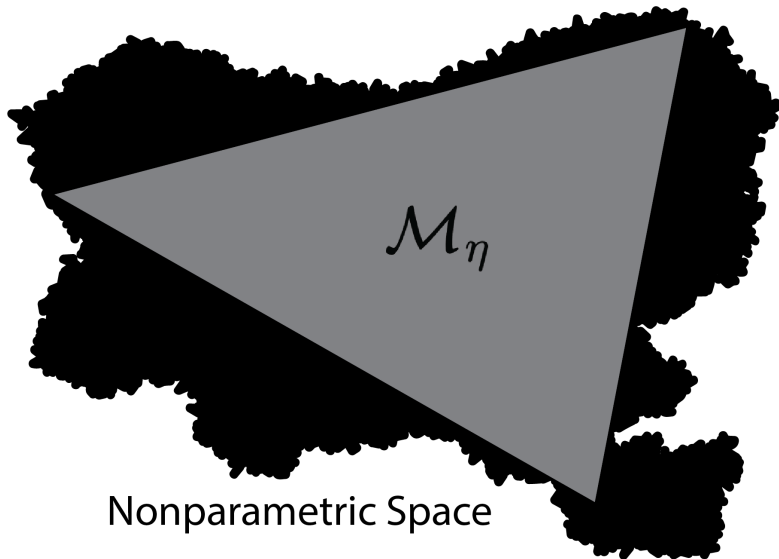
- Model to borrow information from ‘nearby’ observations

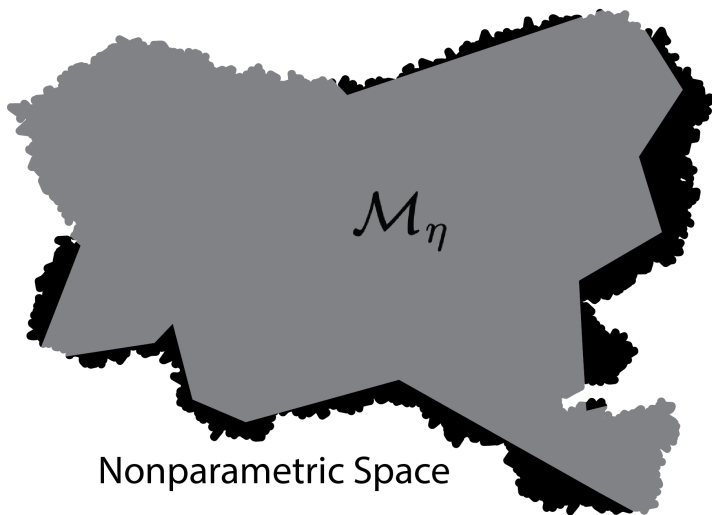
Correct Model Specification

$$\Pr(A = a \mid W) \in \mathcal{M}_\alpha = \{\pi(W; \alpha) : \alpha \in A\}$$

$$E[Y \mid A, W] \in \mathcal{M}_\beta = \{\mu(A, W; \beta) : \beta \in B\}$$

⁴IPW, g-computation, AIPW are exactly equal with saturated models





Challenges for Theory

Wider class does not come freely⁵

1. Convergence rates⁶

- **Solution:** rate-robust estimators (AIPW, TMLE)⁷

$$\left[\frac{1}{\hat{\pi}(W)} - \frac{1}{\pi(W)} \right] \times [\hat{\mu}(A, W) - \mu(A, W)]$$

2. Donsker class restriction⁸

- **Solution:** cross-fitting
- Split into train-test set, fit nuisance models in train set and estimate effect in test set (then swap roles)

⁵Zivich et al. 2022 *Wiley StatsRef*

⁶Daniel 2018 *Wiley StatsRef*

⁷Cost of rate-robustness is double-susceptibility to model misspecification

⁸Chernozhukov et al. 2018 *The Econometrics Journal*

Challenges for Practice

1. Computational complexity
2. Algorithm selection
3. Dependence on Random Number Generators (RNGs)

(Pseudo) Random Number Generators

(Pseudo) Random Number Generators

RNGs are deterministic functions that take a starting state

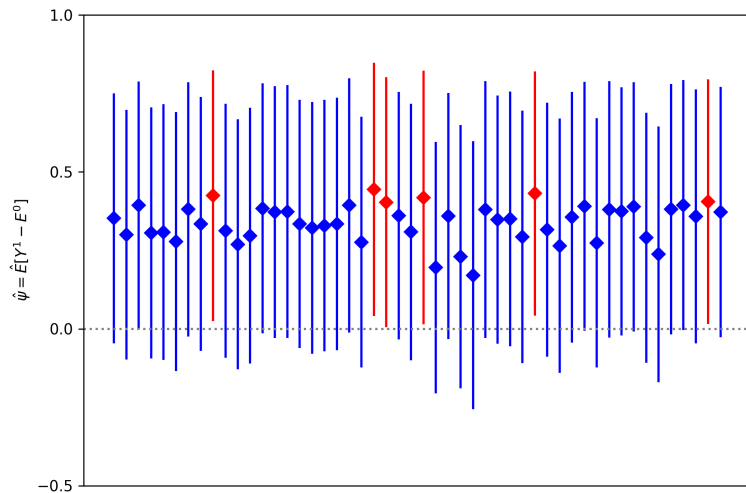
- Seed determines sequence of all generated numbers
- Changing the seed changes the sequence

Role of RNGs in causal effect estimation with machine learning

1. Machine learning algorithms
2. Cross-validated stacking algorithms (e.g., super learner)
3. Data splitting within cross-fitting

Illustration of Seed Dependence

same data set, same estimator, different seeds



Seed Dependence is Unacceptable

Replicability

- Results not reproducible unless seed is known

Offers a new version of p-hacking⁹

- Difficult to uncover

Decision making should not depend on something arbitrary as a choice of seed for a RNG

⁹Some have argued that seeds should be thought of as a hyperparameter that we should tune over (Bethard *arXiv:2210.13393*). I strongly disagree as it is unclear what we are tuning over and this seems ripe for abuse, at least within a publish-or-perish system

Solutions to Seed Dependence¹⁰

Average π and μ predictions under different seeds

- Hard to implement with cross-fitting

Average $\hat{\psi}$ under different seeds

- Easy to implement
- Rubin's Rule for variance

Increase number of folds within cross-fitting

- Applies to cross-fitting or cross-validation

¹⁰Detailed in Schader et al. *Epidemiology* 2024; Zivich *Epidemiology* 2024

Simulations

Data Generating Mechanism¹¹

Propensity score

$$A \sim \text{Bernoulli} \left[0.75 - 1.2 \left(\sum_{j=1}^{100} \frac{\sin(W_i \pi j^3)}{\pi j^3} \right) \right]$$

Outcome model

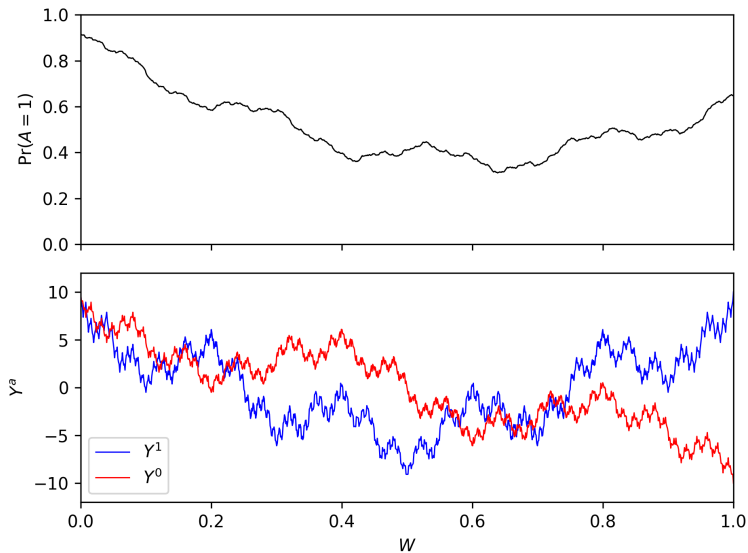
$$Y_i^1 \sim \text{Normal} \left[5 \times \left(\sum_{j=1}^{100} 0.5^j \cos(2W_i 5^j \pi) \right), \sigma \right]$$

$$Y_i^0 \sim \text{Normal} \left[5 \times \left(\sum_{j=1}^{100} 0.5^j \cos(W_i 5^j \pi) \right), \sigma \right]$$

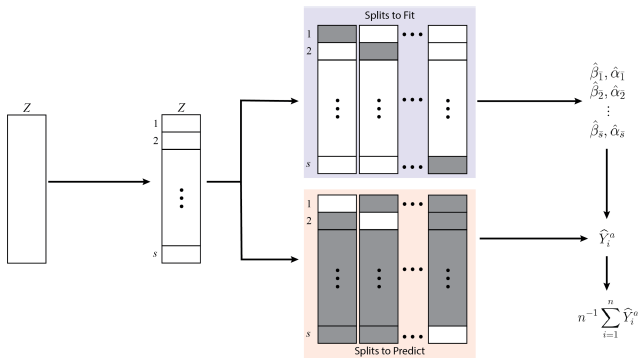
where $\sigma = 1$

¹¹Inspired by Weierstrass's continuous but non-differentiable functions

Data Generating Mechanism



Single Cross-Fit Targeted Maximum Likelihood Estimator



Super learner¹²

- Generalized linear model, cross-validated LASSO, decision tree, 4-layer neural network

¹²Neural network was favored for π . Decision tree was favored for μ

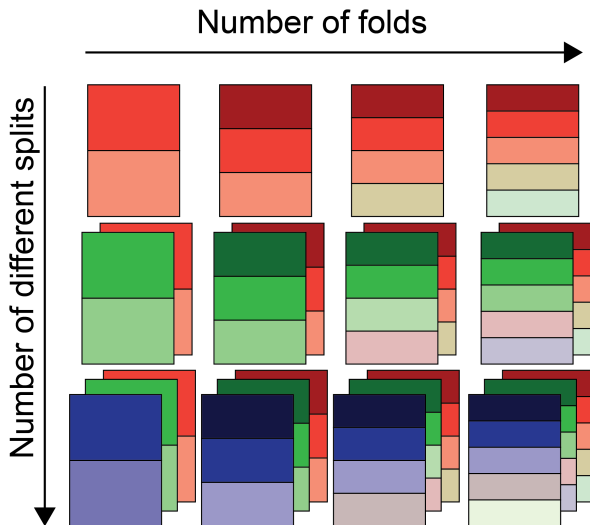
- Generate $n = 1000$ observations
- Apply SCFTMLE under 50 different randomly chosen seeds
- Compute standard deviation (SD) of point and standard error estimates¹³
- Repeat 200 different data sets

Metrics

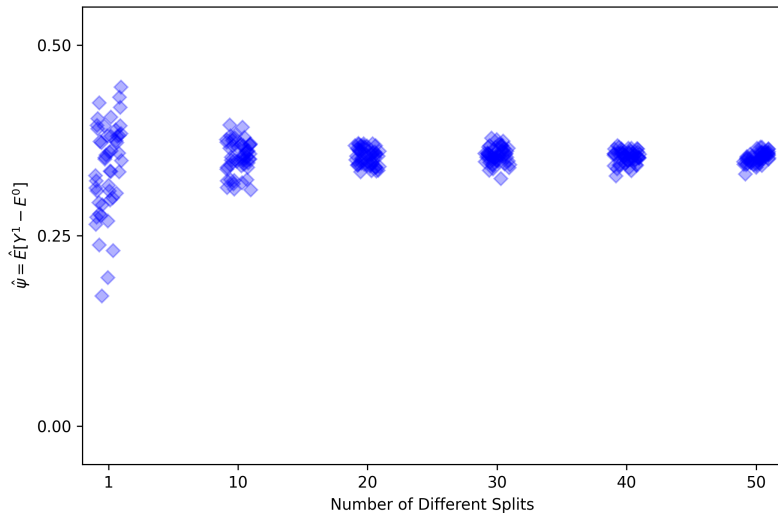
- Mean of SD of point and standard error estimates

¹³Exactly zero for parametric estimators

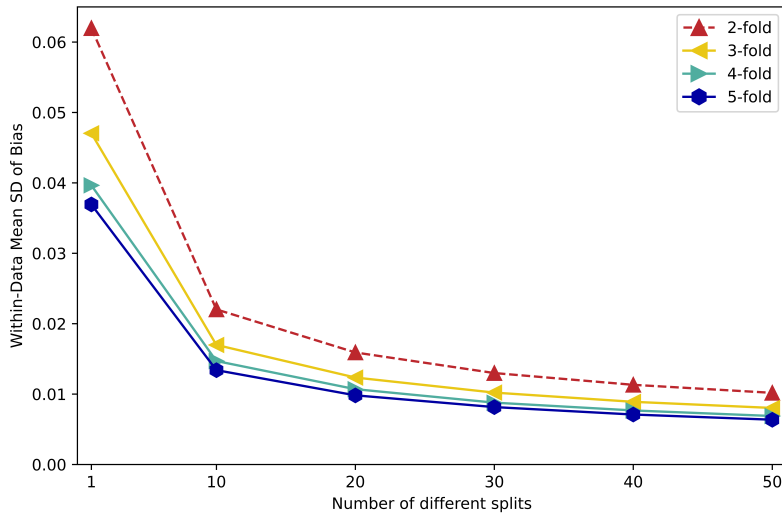
Variations in SCFTMLE



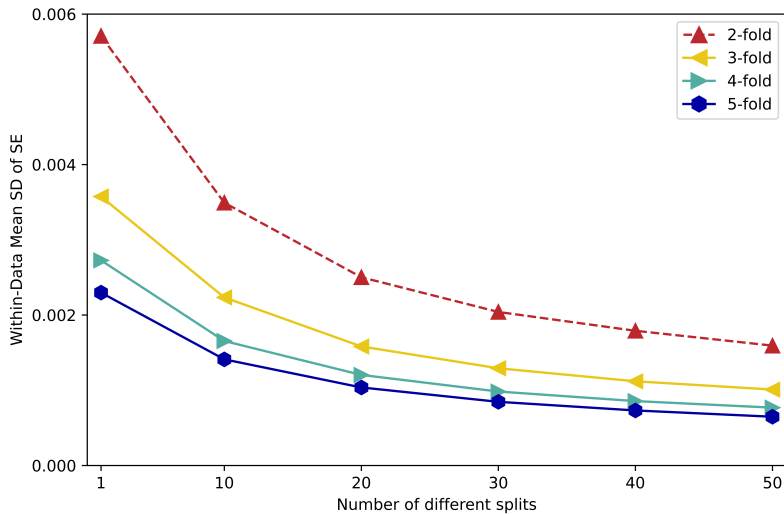
Results – Illustration for Single Data Set



Results – Point Estimate



Results – Standard Error Estimate



Conclusions

RNG seed dependence is unacceptable

Steps can be taken to reduce RNG seed dependence

- Number of splits
- Increasing number of folds
- Easy to implement but computationally intense

Thank you

Questions?



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Zivich “Commentary: The Seedy Side of Causal Effect Estimation with Machine Learning” *Epidemiology* 2024; 35(6):787-790